interneurOn: A Collapsing-Bound Decision Neuron for Neuromorphic Early Exit

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***Abstract*—Digital neuromorphic systems often lack context-aware runtime control, so spiking workloads are commonly run to a fixed horizon even when completion time is input-dependent. We propose interneurOn, a single spike-domain neuron for local early exit without an external controller. At each time step, it updates a persistent evidence state from per-channel inputs, forms a locked signed top-2 margin, and integrates it into a decision variable tested against a collapsing bound. We evaluate interneurOn on radar activity recognition, spoken-digit classification, and a synthetic graph/wavefront benchmark. We also realise it as a fixed-point SpiNNaker-1 kernel in two radar deployments: one measuring accuracy–energy trade-offs and one analysing the effect of upstream evidence encoding on early exit. The neuron uses only small local state, has bounded computational cost per update, and achieves competitive early-exit performance while remaining feasible under neuromorphic hardware constraints.**

***Index Terms*—adaptive inference, early exit, evidence accumulation, spiking neural networks, neuromorphic hardware, SpiNNaker**

I. Introduction

In digital neuromorphic systems, the time to reach a reliable decision is input-dependent, but inference is commonly run to a fixed horizon. As a result, computation may continue after sufficient evidence has already accumulated. Early exit can reduce this excess, but in many spiking early-exit approaches the stop decision is still implemented through a separate monitoring mechanism that reads confidence or output statistics, adding coordination and routing overhead.

For neuromorphic hardware, the stopping question is not only when to stop, but how to realise stopping in the spike domain with small local state, bounded per-update cost, and no global control traffic. This motivates using biological decision models and their circuit-level abstractions as a basis for hardware-compatible stopping.

In this paper, we introduce interneurOn, a single spike-domain neuron that converts local channel activity into a halt spike. It updates a persistent evidence state, forms a locked signed top-2 margin, and integrates this margin into a decision variable. The neuron emits one halt spike per window, either on early commitment or at the terminal step. Because the stopping decision is made locally, the mechanism avoids a separate supervisory pathway and fits the event-driven execution model of neuromorphic hardware. Fig. 1 contrasts this local stopping mechanism with controller-based external stopping, where halting is triggered by confidence or summary statistics.

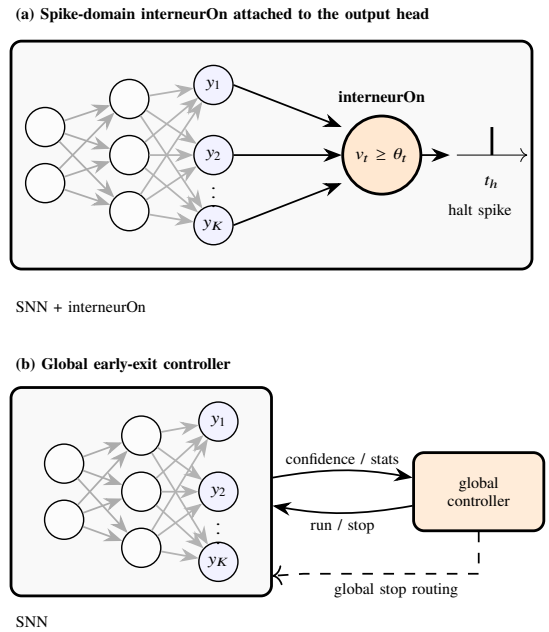

Fig. 1. Early-exit mechanisms in the spike domain. **(a)** interneurOn locally reads activity from the output channels and emits a halt spike. **(b)** A SEENN-style design [4] uses an external controller to issue network-wide stop events.

Contributions:

1) A controller-free local early-exit primitive, interneurOn, that maps local evidence to a halt spike under a collapsing-bound rule with forced halting at the terminal step.
2) A locked signed top-2 margin stopping rule in the spike domain, with lock-on, lock-out, dwell, and symmetric negative reset to improve robustness under unstable winner dynamics.
3) A cross-domain evaluation against early-exit baselines on the Intelligent RF Sensing for Falls and Health Prediction (INSHEP) radar dataset, a subset of the Spiking Heidelberg Digits (SHD) dataset, and a synthetic graph/wavefront stopping benchmark, including two SpiNNaker-1 radar deployments that measure accuracy–energy trade-offs and analyse the effect of upstream evidence encoding on early exit.

II. Background and Related Work

Early exit in deep and spiking networks: Early exit in deep networks uses intermediate heads and adaptive halting rules, including patience-based stopping [1]–[3]. In spiking networks, temporal early exit monitors spike-domain confidence or stability [4], [5], often via an external controller.

In contrast, we realise stopping directly in the spike domain: a single neuron emits a halt spike when its decision variable crosses a time-varying bound. This connects adaptive early exit to ideas from sequential evidence accumulation in a form suitable for digital neuromorphic systems. More broadly, this relates to earlier notions of self-delimiting neural computation [6]: in neuromorphic systems, computation unfolds in time through events, so stopping can arise from the dynamics rather than from a separate synchronised controller.

Decision bounds and evidence accumulation: Diffusion and accumulator models describe stopping as evidence integration toward a threshold [7]. When delaying a decision has a cost, collapsing thresholds can be optimal or near-optimal [8]. Classic sequential baselines include MSPRT and leaky competing accumulators [9], [10]. interneurOn follows the same general principle, but instantiates it through a local locked top-2 margin and a collapsing bound in the spike domain.

Neuromorphic graph-style computations: Some non-learning neuromorphic workloads can be expressed as graph dynamics in which spikes implement local message passing and completion is tied to an arrival event. Examples include spiking wavefront propagation and related shortest-path primitives [11], [12]. To study stopping in this setting, we use a synthetic graph/wavefront benchmark with ground-truth arrival ordering.

Hardware context and neuromorphic constraints: Neuromorphic platforms emphasise local memory, sparse event routing, and low computational cost per update [13], [14]. Controller-based stopping can introduce additional routing and coordination overhead. interneurOn instead uses on-core state and emits a single halt spike, which can be combined into a network-wide stop signal via existing multicast routing.

III. Method

A. Definitions and setup

a) Discrete-time updates.: We define the stopping rule in discrete time indexed by t. Task-specific preprocessing maps each input stream to a sequence of algorithmic updates. On SpiNNaker, the kernel is executed once per algorithmic update, i.e., once per application of the state-transition rule. We use "time step" for the algorithmic dynamics and report hardware cost separately in cycles per update.

b) Budget and hypotheses.: We use budget to denote the resource traded against accuracy at decision. Depending on the setting, budget may correspond to elapsed time, executed time steps, or processed events. A hypothesis denotes one competing alternative (e.g. a class label or candidate source). Each hypothesis is represented by one local channel $k \in \{1, \ldots, K\}$.

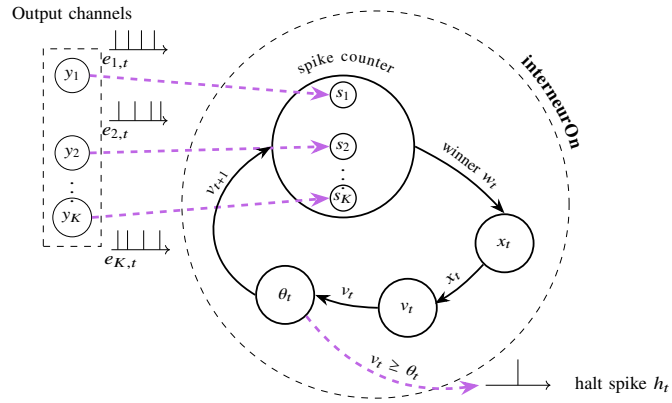

Fig. 2. Spike-domain signal path of interneurOn. Per-channel inputs $e_{k,t}$ from the competing output channels update the evidence state $s_{k,t}$ (Eq. (1)); the resulting margin drives the decision variable, and halting occurs once $v_t \geq \theta_t$ holds for k_{dwell} consecutive time steps.

c) Overview of the stopping rule.: interneurOn integrates a locked signed top-2 margin against a collapsing bound to determine halting in discrete time. Lock-on selects and locks the currently leading hypothesis for comparison. Lock-out and dwell reduce premature commitment under brief winner fluctuations. Symmetric negative reset clears the decision state after strong contradictory evidence so accumulation can restart under a new winner. The stopping rule is fully deterministic for fixed inputs, parameters, and initial state. Unlike classical stochastic evidence-accumulation models, any variability in halting arises from variability in the input evidence stream rather than from randomness in the decision mechanism itself.

B. System model and evidence interface

interneurOn receives a K-channel evidence stream over a window of T discrete time steps $t = 0, \ldots, T-1$, with one channel per competing hypothesis. At each step it receives per-channel inputs $e_t = \{e_{k,t}\}_{k=1}^{K}$ and updates a persistent evidence state $s_t = \{s_{k,t}\}_{k=1}^{K}$, from which the winner, margin, and halting decision are derived (Fig. 2).

Throughout, $e_{k,t}$ denotes the task-specific per-channel evidence input at time step t. Depending on the setting, it may be a spike count, weighted count, activation, or calibrated posterior. The same formulation also applies in the limiting case of winner-take-all outputs, although the available evidence is then much sparser.

C. Decision dynamics

1) Evidence-state update:

$$s_{k,t} = (1-\eta)\, s_{k,t-1} + e_{k,t}, \qquad t = 0, \ldots, T-1, \tag{1}$$

with initial condition $s_{k,-1} = 0$ and $\eta \in [0, 1]$. Here $e_{k,t}$ is the per-step evidence input. With $\eta = 1$, the state is instantaneous ($s_{k,t} = e_{k,t}$); with $\eta < 1$, it becomes an exponentially weighted sum over past steps.

2) Winner selection and locked signed top-2 margin: All competition is defined on $s_{k,t}$. The instantaneous winner is

$$w_t = \arg\max_k s_{k,t}. \tag{2}$$

If multiple channels attain the maximum, ties are broken by selecting the smallest channel index.

After lock-on, the neuron compares the locked hypothesis $c^\star$ only against its strongest rival:

$$s_{(1),t} = s_{c^\star,t}, \qquad s_{(2),t} = \max_{j \neq c^\star} s_{j,t}, \tag{3}$$

forming the signed locked top-2 margin

$$x_t = s_{(1),t} - s_{(2),t}. \tag{4}$$

This margin is the evidence signal used by the accumulator.

3) Leaky accumulation and collapsing bound: The neuron integrates x_t into a scalar decision variable

$$v_t = (1-\lambda)\, v_{t-1} + g\, x_t, \qquad t = 0, \ldots, T-1, \quad v_{-1} = 0, \tag{5}$$

with leak $\lambda \in [0, 1)$ and gain $g > 0$. Halting is tested against the collapsing bound

$$\theta_t = \max\!\big(\theta_{\min},\, b_0 - \beta\, t\big), \tag{6}$$

where b_0, β, and $\theta_{\min}$ encode the accuracy–latency–energy trade-off.

4) Decision rule and stabilisation mechanisms: In addition to (v_t, θ_t), the neuron maintains small discrete state: previous winner w_{t-1}, locked hypothesis $c^\star$, dwell counter d, lock-out counter ℓ, and a halt flag.

We initialise $v_{-1} = 0$, $c^\star = \varnothing$, $d = 0$, $\ell = 0$, and `halted=false`.

After emitting an early halt spike, the neuron ignores subsequent inputs for the remainder of the window. If no early halt occurs, a *forced halt* is emitted at the terminal step $t = T-1$. At halt time t_h (early or forced), we output $\hat{y} = c^\star$; if $c^\star = \varnothing$, we fall back to $\hat{y} = \arg\max_k s_{k,t_h}$.

a) Lock-on and flip shielding.: If no hypothesis is locked, we set $c^\star \leftarrow w_t$. A flip occurs when $w_t \neq w_{t-1}$; in that case, we re-lock to the new winner and reset dwell by setting $c^\star \leftarrow w_t$ and $d \leftarrow 0$. If lock-out is enabled ($L > 0$), a winner flip starts a lock-out phase. During subsequent updates with $\ell > 0$, the accumulator is frozen, the dwell counter is reset, the previous-winner index is updated, and the lock-out counter is decremented. The exact update is given in Alg. 1.

b) Symmetric negative reset.: When enabled, a large negative excursion resets the state:

$$v_t < -\theta_t \Rightarrow v_t \leftarrow 0, \quad d \leftarrow 0, \quad c^\star \leftarrow \varnothing. \tag{7}$$

c) Dwell-based halting.: Halting requires persistence above threshold for k_{dwell} consecutive time steps:

$$d \leftarrow \begin{cases} d+1, & v_t \geq \theta_t, \\ 0, & v_t < \theta_t, \end{cases} \qquad d \geq k_{\text{dwell}} \Rightarrow \text{emit one halt spike.} \tag{8}$$

If no early halt occurs, we emit a forced halt at $t = T-1$ using the same readout rule above.

Fig. 3 illustrates the evidence states, the locked signed top-2 margin, and the collapsing-bound response.

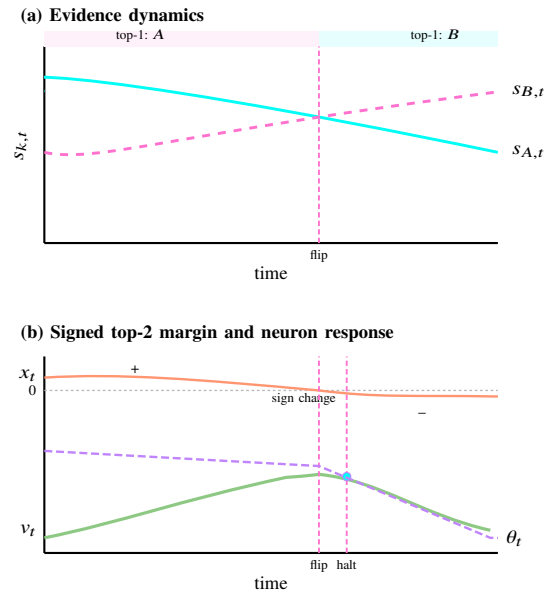

Fig. 3. Illustration of the decision dynamics. **(a)** Two example evidence states $s_{A,t}$ and $s_{B,t}$ intersect at a winner flip. **(b)** Schematic evolution of the signed top-2 margin x_t, decision variable v_t, and collapsing bound θ_t; halting occurs once v_t crosses θ_t with sufficient dwell.

D. Hardware realisation

We implement the stopping rule as a fixed-point SpiNNaker-1 kernel with bounded per-update cost. At each on-chip update, channel counts $e_{k,t}$ are incorporated into the persistent evidence state used for winner selection and competition. Alg. 1 gives the kernel update, and Table I reports the memory footprint and measured cycle cost. In our SpiNNaker experiments, one on-chip update is scheduled every 1 ms.

a) Channel interface (how $e_{k,t}$ is provided).: Each channel k corresponds to one hypothesis, and at update t interneurOn receives the count $e_{k,t}$. In the radar deployments, these counts come either from replayed class-channel spikes (dense) or from output spikes of the on-chip readout population (sparse), accumulated into local DTCM counters and cleared after each update.

b) Local state (what is per-channel vs. scalar).: All state is stored locally on the core: per-channel evidence state $\{s_k\}_{k=1}^{K}$ and scalar state consisting of decision variable v, previous winner index w_{prev}, locked hypothesis index $c^\star$ (or unset), dwell counter d, lock-out counter ℓ, and a halted flag; the bound θ_t is computed each update from $(b_0, \beta, \theta_{\min})$. Only $\{s_k\}$ scales with K; w_{prev} and $c^\star$ are single indices (not per-channel).

c) Compute and output.: Compute scales linearly with K (a fixed number of fixed-point operations per channel plus a small constant-cost state machine). When the dwell criterion is satisfied, the kernel emits exactly one halt spike and suppresses further processing for the remainder of the window; if no early halt occurs by $t = T-1$, a forced halt is issued.

IV. Experimental Setup

We first define the common operating-point and evaluation protocol, then describe the radar, SHD, and synthetic

Algorithm 1 Per-update kernel for interneurOn.

Input: time step index t, window length T, channel evidence inputs $e_{1:K,t}$
State: evidence $s_{1:K}$, decision variable v, locked hypothesis $c^\star$, previous winner w_{prev}, dwell counter d, lock-out counter ℓ, halt flag `halted`
Params: gain g, leaks (λ, η), bound schedule $(b_0, \beta, \theta_{\min})$, lock-out length L, dwell length k_{dwell}, negative-reset flag
if `halted` **then** ▷ one-shot halt
  return
end if
$\theta_t \leftarrow \max(\theta_{\min}, b_0 - \beta t)$ ▷ collapsing bound
for $k = 1, \ldots, K$ **do**
  $s_k \leftarrow (1-\eta)s_k + e_{k,t}$ ▷ evidence-state update, Eq. (1)
end for
$w_t \leftarrow \arg\max_k s_k$ ▷ instantaneous winner, Eq. (2)
if $\ell > 0$ **then** ▷ lock-out: freeze integration
  $\ell \leftarrow \ell - 1$
  $d \leftarrow 0$
  $w_{\text{prev}} \leftarrow w_t$
  return
else if $c^\star = \varnothing$ **then** ▷ lock-on
  $c^\star \leftarrow w_t$
else if $w_t \neq w_{\text{prev}}$ **then** ▷ winner flip
  $c^\star \leftarrow w_t$ ▷ re-lock
  $d \leftarrow 0$
  if $L > 0$ **then**
    $\ell \leftarrow L$ ▷ start lock-out
    $w_{\text{prev}} \leftarrow w_t$
    return
  end if
end if
$w_{\text{prev}} \leftarrow w_t$
$x_t \leftarrow s_{c^\star} - \max_{j \neq c^\star} s_j$ ▷ locked top-2 margin, Eq. (4)
$v \leftarrow (1-\lambda)v + g\, x_t$ ▷ leaky accumulation, Eq. (5)
if `neg-reset` and $v < -\theta_t$ **then** ▷ symmetric negative reset, Eq. (7)
  $v \leftarrow 0$
  $d \leftarrow 0$
  $c^\star \leftarrow \varnothing$
  return
else if $v \geq \theta_t$ **then** ▷ dwell update
  $d \leftarrow d + 1$
else
  $d \leftarrow 0$
end if
if $d \geq k_{\text{dwell}}$ **then** ▷ early commitment
  emit halt spike
  `halted` $\leftarrow$ `true`
  return
end if
if $t = T - 1$ **then** ▷ forced halt
  emit forced halt spike ▷ apply the halt-time readout rule
  `halted` $\leftarrow$ `true`
end if

TABLE I
RUNTIME AND MEMORY FOOTPRINT OF INTERNEURON ON SPINNAKER-1 (200 MHZ, 1 MS TIMER INTERVAL, 64 NEURONS PER CORE, K=6 CLASS CHANNELS FOR INSHEP). REPORTED CYCLES ARE CORE CLOCK CYCLES PER UPDATE. STATE MEMORY INCLUDES K-CHANNEL EVIDENCE.

Item	Value	Notes
Bytes of parameters	40 B	per neuron (S16.15 fixed-point)
Bytes of state	96 B	per neuron (DTCM)
Cycles per update (core)	33 974	measured mean over epoch
Cycles per update (neuron)	531	33 974 / 64
Core utilisation (per update)	17.0 %	33 974 / 200 000

graph/wavefront experiments. Posterior-space experiments isolate the stopping rule before testing the same mechanism in spike/count form on SpiNNaker.

A. Common operating-point and evaluation protocol

We define an operating point (halting parameters, not classifier weights) by

$$\Theta = (g, \lambda, b_0, \beta, \theta_{\min}, k_{\text{dwell}}, L, \text{neg-reset}, \eta),$$

which fully specifies the per-time-step update in Sec. III-C. We refer to $(b_0, \beta, \theta_{\min})$ as the budget schedule because it sets the time-varying bound θ_t. Within each task we keep λ, the budget schedule, k_{dwell}, L, neg-reset, and (when used) η fixed, and use g as the main knob that shifts the decision-time distribution.

We denote decision time by T^* and summarise it by the median $t_{50} = \text{median}(T^*)$, which serves as the time-budget proxy for radar and SHD. In radar, operating points are selected directly on R3 to obtain matched achieved medians $t_{50} \in \{9, 12, 15\}$ for fair trade-off comparison across stopping rules; when exact matching is not possible, we report the nearest achievable point. In SHD, operating points are tuned on a validation split and transferred unchanged to the test split, where we report achieved t_{50}, accuracy at decision, mean top-1 flips, and decision-time IQR.

We report decision time T^* in processed-step units, so for a window of length T the maximum reported decision time is T, although the last update index is $T - 1$.

For the synthetic graph benchmark, methods are matched by target mean event budget B (Eq. (12)) rather than by time steps, because event density varies across trials without a proportional change in elapsed time. We report the achieved mean event count for the nearest feasible operating point.

B. Radar experiments

Radar classification (INSHEP).: We evaluate on the INSHEP radar dataset [15], comprising six human activities (walking, sitting, standing, pick-up, drinking, simulated fall) recorded with a 5.8 GHz FMCW radar (400 MHz bandwidth) across five collection campaigns. We train and calibrate on repetitions R1–R2 and test on repetition R3. Each recording is converted into a prefix stream of length T=20 as more radar chirps become available. Each prefix is mapped to a 51-dimensional feature vector combining temporal energy descriptors with pooled Range–Doppler features (K_r=5 dominant range bins, B_d=24 Doppler bands). Features are globally z-scored before entering the classifiers.

Posterior-space (CPU).: For the radar posterior-space experiment, we train a one-hidden-layer 256-unit ReLU MLP on R1–R2 using the prefix features above and apply temperature scaling on the same split to obtain calibrated posterior streams $p_{k,t}$. Evaluation is on the held-out repetition split R3. At the full 20-step window, the classifier reaches 0.765 top-1 and 0.960 top-3 accuracy on R3. As an upper-bound reference on prefix-wise separability, an oracle-over-time metric that retrospectively selects the best prefix per stream reaches 0.928 top-1. These posterior streams are used by the posterior-space baselines and by interneurOn-CPU.

Dense evidence replay on SpiNNaker (interneurOn-SPINN-Dense).: For the dense deployment, we keep the upstream radar classifier on the host and replay only its per-time-step class logits $z_{k,t}$ on SpiNNaker, yielding a controlled class-evidence interface for hardware transfer. We transform these logits into non-negative evidence

$$E_{k,t} = z_{k,t} - \min_j z_{j,t} + \delta, \tag{9}$$

with $\delta > 0$. Each $E_{k,t}$ is rounded to an integer and clipped to $[0, 2]$, yielding $n_{k,t} \in \{0, 1, 2\}$. At replay step t, exactly $n_{k,t}$ spikes are injected on class channel k. One replay step corresponds to one on-chip update (1 ms), so the effective per-step input to interneurOn is $e_{k,t} = n_{k,t}$.

Sparse on-chip linear readout on SpiNNaker (interneurOn-SPINN-Sparse).: As a lower-rate on-chip readout path, we use a separate linear readout on the same prefix features F_t, realised in spike form on SpiNNaker. On the host we fit weights W and bias b for the linear score

$$z_t = WF_t + b. \tag{10}$$

At the full 20-step window on R3, this linear model reaches 0.605 top-1 and 0.893 top-3 accuracy; its oracle-over-time top-1 accuracy is 0.853. On chip, we do not implement the full affine map exactly: only the signed weighted-sum term WF_t is realised through spike-based excitatory and inhibitory drive onto a class readout population, and the predicted label is decoded from the resulting output spike counts. The bias term b is omitted.

Signed feature encoding and interface to interneurOn.: Each feature dimension F_d is split into two non-negative channels,

$$F_d^+ = \max(F_d, 0), \qquad F_d^- = \max(-F_d, 0), \tag{11}$$

mapped to indices $2d$ and $2d+1$. These signed channels are encoded as spike trains and projected to the class readout population with excitatory and inhibitory synapses according to the sign of $W_{k,d}$. The resulting class activity approximates the signed linear score in spike form, with bias omitted. InterneurOn receives only the per-time-step class counts $e_{k,t}$ derived from these readout spike trains.

Encoding variants (evidence-timing sweep).: To study how upstream spike timing affects early exit under the same count-based interface, we generate several Poisson encodings of the signed feature channels, run the on-chip linear readout for T_{hw}=20 time steps, and record the resulting class spike trains. The variants E1–E4 in Table V differ only in the global rate scale and per-time-step spike cap, which together determine how much class evidence appears early versus late in the window. We summarise the resulting class-spike profile by Early5, Early8, and Total. A rigid TTFS-style variant (at most one spike per signed-feature channel over the full window) produced too little downstream class evidence for reliable calibration and is therefore omitted.

Posterior-space baselines.: For radar, in the CPU posterior-space setting and using posteriors p_t, we compare against Max-Prob, Entropy, Top-2 log-odds, and Race/LCA. Max-Prob halts when $\max_k p_{k,t} \geq \tau$; Entropy halts when $H(p_t) \leq \tau$; Top-2 log-odds uses an MSPRT-like cumulative log-odds test on the top-2 classes with a fixed threshold; and Race/LCA uses leaky competing accumulators and a margin/dwell criterion.

C. SHD posterior-space experiment

We use a 10-class subset of the Spiking Heidelberg Digits (SHD) dataset [16] (classes "zero" through "nine"). Spike times are binned to 1 ms, clipped at 200 ms, and pooled into B_s=16 channels. The stream is grouped into T=20 time steps of 10 ms. For each time step we compute simple activity descriptors, stack the last K_τ=3 steps, and feed them to a 128-unit ReLU MLP. After isotonic calibration, the resulting posterior stream $P_t \in \mathbb{R}^{10}$ is used by all SHD early-exit rules. SHD is evaluated only in posterior space, with operating points tuned on a validation split and transferred to the test split.

Posterior-space baselines.: We use the same posterior-space baselines as for radar: Max-Prob, Entropy, Top-2 log-odds, and Race/LCA.

D. Synthetic graph/wavefront benchmark

We evaluate early exit in a synthetic event/count setting inspired by spiking wavefront navigation and neuromorphic shortest-path formulations [11], [12]. Each trial is defined on a sparse planar graph derived from a Voronoi skeleton, with multiple candidate sources and one sink. Hidden source-to-sink travel times determine the true earliest-arrival label. Methods observe only noisy online sink counts $e_{k,t} \in \mathbb{N}$ generated by these hidden arrivals and must infer the earliest-arrival source from partial evidence evolving over time, without access to the graph, edge delays, or arrival times.

For this benchmark we compare methods by processed-event budget rather than elapsed time:

$$B = \mathbb{E}\left[\sum_{t=0}^{T^*-1} \sum_{k=1}^{M} e_{k,t}\right], \tag{12}$$

where T^* is the decision time. Under an event-driven cost model, dynamic work scales approximately with the number of processed events up to decision, so B serves as an energy proxy rather than an exact hardware energy model.

To avoid operating points that match the target mainly by forced halts at the terminal step, we restrict matching to parameters whose miss rate does not exceed 0.10.

Baselines.: For the graph benchmark (event/count), using sink counts $e_{k,t}$, we compare against CumGap, Poisson-LLR, Race-fixed, an arrival-time oracle, and a fixed-time rule. CumGap accumulates raw per-channel counts and halts when the cumulative top-2 gap exceeds a fixed margin for k_{dwell} time steps; Poisson-LLR forms cumulative per-channel log-likelihood-ratio scores under a simple Poisson count model and halts when the top-2 score gap exceeds a fixed threshold for k_{dwell} time steps; Race-fixed uses leaky per-channel traces and halts when the top-2 gap exceeds a fixed margin for k_{dwell} time steps; the arrival-time oracle (reference only) halts at $t_{\text{true}} + \Delta$, where $t_{\text{true}} = \min_k t_k$ is known only for evaluation; and Fixed-time halts at $T_{\max}$.

V. Results

We report posterior-space results on radar and SHD, then the SpiNNaker radar deployments, the synthetic graph/wavefront benchmark, and finally ablations and per-stream examples.

TABLE II

INSHEP POSTERIOR-SPACE METHOD COMPARISON (N=375, T=20, K=6) AT TARGET MEDIAN DECISION TIMES $t_{50} \in \{9, 12, 15\}$.

	target t_{50} = 9			target t_{50} = 12			target t_{50} = 15		
Method	Acc	Flips	IQR	Acc	Flips	IQR	Acc	Flips	IQR
Max-Prob	0.70	2.06	3–20	0.72	2.15	4.5–20	0.72	2.18	5–20
Entropy	0.70	2.06	3–20	0.72	2.15	5–20	0.74† [16]	2.20	6–20
Top-2 log-odds	0.77	2.12	6–15	0.78	2.25	8–20	0.78† [13]	2.26	8.5–20
Race/LCA	0.77	2.04	6–14	0.79	2.13	8–16.5	0.78	2.28	10–20
interneurOn-CPU	**0.78**	**1.99**	8–12	**0.81**	2.13	10–15	**0.79**	**2.24**	13–17

† nearest-match operating points; brackets show the achieved median.

TABLE III

SHD POSTERIOR-SPACE METHOD COMPARISON (N=1079, T=20, K=10), WITH VALIDATION-TUNED OPERATING POINTS. ACH. t_{50} DENOTES THE ACHIEVED TEST-SET MEDIAN DECISION TIME.

	target t_{50} = 9				target t_{50} = 12				target t_{50} = 15			
Method	Ach. t_{50}	Acc	Flips	IQR	Ach. t_{50}	Acc	Flips	IQR	Ach. t_{50}	Acc	Flips	IQR
Max-Prob	8.0	0.56	4.90	5–13	11.0	0.72	5.82	6–17	16.0	**0.78**	6.66	8–20
Entropy	8.0	0.58	5.07	5–13	13.0	**0.75**	6.17	7–19	15.0	**0.78**	6.63	8–20
Top-2 log-odds	8.0	0.57	4.71	6–12	11.0	0.71	5.75	7–16	15.0	0.77	6.56	10–20
Race/LCA	8.0	**0.61**	5.15	7–13	12.0	**0.73**	6.10	9–17	15.0	0.75	6.62	10–20
interneurOn-CPU	8.0	0.60	4.90	6–12	12.0	**0.73**	5.91	8–16	14.0	**0.77**	6.49	10–19

Tables II and III isolate sequential halting on calibrated posteriors. On INSHEP radar (Table II), interneurOn-CPU attains the highest reported accuracy at all three comparison points. It also gives tighter decision-time distributions than Max-Prob and Entropy, and at low budget shows the lowest mean number of top-1 flips.

On SHD (Table III), validation-tuned operating points transfer imperfectly to the test set, so achieved medians differ from their nominal targets. interneurOn-CPU remains competitive but is not uniformly best: Race/LCA is strongest at the lowest achieved budget, while Entropy and Max-Prob attain the highest accuracies at the largest achieved budgets.

A. Radar deployments on SpiNNaker

1) Dense evidence replay and gain calibration on SpiNNaker (interneurOn-SPINN-Dense): For interneurOn-SPINN-Dense, we fix $b_0 = 0.4$, $\beta = 0.02$, $\theta_{\min} = 0.05$, $\lambda = 0.05$, $L = 1$, and $k_{\text{dwell}} = 1$, with two-sided negative reset. For each target median halting time $t_{50}^{\text{target}} \in \{2, 4, \ldots, 20\}$, we first sweep g in a CPU count-based mirror of interneurOn using the same counts $e_{k,t} = n_{k,t}$, and then refine the operating point on hardware by a small empirical sweep around the corresponding gain.

Table IV shows that the gain-to-median mapping remains monotonic on hardware and broadly follows the CPU count-based calibration sweep, with small deviations due to event delivery and halt readout on SpiNNaker.

Injected event load.: Across the N=375 streams with T_{hw}=20, dense evidence replay produces 16 043 injected spikes in total, capped at 2 per class channel per time step.

2) On-chip energy–accuracy trade-off: To characterise the energy–accuracy trade-off of interneurOn-SPINN-Dense on INSHEP, we reuse the calibrated gains g from Sec. V-A1. For each calibrated gain, we clamp the hardware runtime horizon to $T_{\text{run}} \in \{2, 4, \ldots, 20\}$ time steps, run one epoch over all N=375 streams, and read the dynamic execution energy from SpiNNaker's energy report.

TABLE IV

CPU-INITIALISED AND HARDWARE-REFINED OPERATING POINTS FOR INTERNEURON-SPINN-DENSE ON INSHEP (N=375). FOR EACH TARGET MEDIAN HALTING TIME, g_{CPU} IS SELECTED FROM THE CPU COUNT-BASED SWEEP AND g_{HW} IS THE FINAL HARDWARE-REFINED GAIN USED TO MATCH THE REALISED SPINNAKER MEDIAN.

target	g_{CPU}	g_{HW}	t_{50}^{CPU}	acc_{CPU}	t_{50}^{HW}	acc_{HW}
2	0.207	0.210	2	0.597	2	0.602
4	0.125	0.190	4	0.677	4	0.688
6	0.078	0.120	6	0.723	6	0.723
8	0.059	0.070	8	0.733	8	0.765
10	0.034	0.050	10	0.752	10	0.760
12	0.022	0.035	12	0.768	12	0.771
14	0.016	0.020	14	0.747	14	0.784
16	0.011	0.015	16	0.733	16	0.771
18	0.006	0.010	18	0.715	18	0.763
20	0.004	0.001	20	0.691	20	0.747

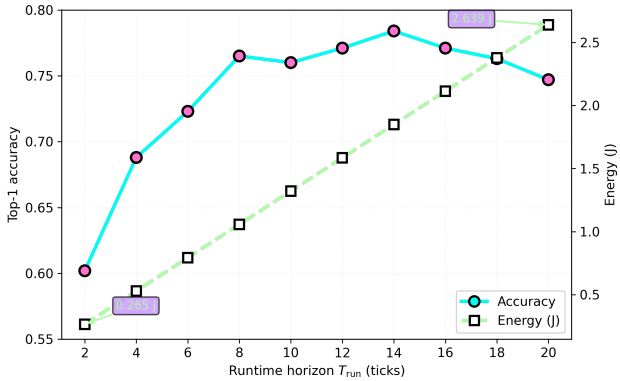

Fig. 4. Accuracy at decision versus measured SpiNNaker-1 execution energy for interneurOn-SPINN-Dense on INSHEP (N = 375). Each point is obtained by clamping the runtime horizon to $T_{\text{run}} \in \{2, 4, \ldots, 20\}$ time steps (same mapping and kernel), running one epoch over all streams, and reading "simulation execution energy (active chips and cores only)" from the SpiNNaker energy report.

Figure 4 shows the resulting accuracy–energy trade-off.

We observe a knee around $t_{50} \approx 12$, beyond which additional runtime mainly increases energy cost with little further benefit and a slight accuracy drop at the highest budgets.

3) Sparse on-chip classifier (interneurOn-SPINN-Sparse): We next evaluate interneurOn-SPINN-Sparse, in which the class-count evidence stream $e_{k,t}$ is generated on chip by a spike-based linear readout rather than replayed from host-computed logits.

For this sparse deployment we use the more permissive bound schedule

$$b_0 = 2.0,\ \beta = 0.05,\ \theta_{\min} = 0.5,\ \lambda = 0.05,\ L = 0,\ k_{\text{dwell}} = 1,$$

and sweep g to match target medians.

a) Sparse replay under controlled class-spike statistics (on-chip).: Table V summarises the class-output spike statistics for the four encoding variants (E1–E4). For each variant, we sweep g to approximately match target medians $t_{50} \in$

TABLE V

SIGNED-FEATURE ENCODING VARIANTS USED TO GENERATE THE SPIKE INPUT TO THE SPARSE ON-CHIP LINEAR HEAD AND THE RESULTING CLASS SPIKE TRAINS USED AS INPUT TO INTERNEURON. "EARLY5" AND "EARLY8" DENOTE THE SUM OVER CLASSES OF CLASS SPIKES EMITTED IN TIME STEPS $t \in [0, 4]$ AND $t \in [0, 7]$, RESPECTIVELY; "TOTAL" SUMS OVER $t \in [0, 19]$.

ID	Encoding	Rate scale	Cap/tick	Early5	Early8	Total
E1	Poisson (baseline)	500	1	1	91	1145
E2	Poisson (late-heavy)	800	1	1	198	1685
E3	Poisson (early-active)	500	3	14	236	1747
E4	Poisson (high-rate)	1500	1	11	363	2322

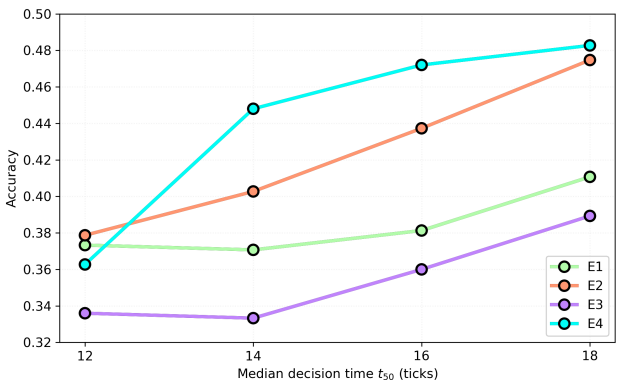

Fig. 5. Sparse on-chip replay accuracy at decision versus matched median halting time t_{50} for the four encoding variants E1–E4 (Table V).

$\{12, 14, 16, 18\}$, measure T^* from the on-chip halt spike, and decode the label from cumulative class-output spike counts up to T^*. Fig. 5 plots accuracy at decision versus achieved median decision time.

Across encodings, accuracy improves with budget, but the trade-off depends strongly on the temporal profile of the class evidence. The high-rate variant E4 gives the strongest overall trade-off, reaching 0.448 at $t_{50} = 14$ and 0.483 at $t_{50} = 18$. E1 and E2 improve more gradually, with E2 approaching 0.475 at $t_{50} = 18$. Despite a larger EARLY5 spike count, E3 underperforms, indicating that early spike quantity alone is insufficient if it does not produce a reliable top-2 margin after the linear readout.

B. Synthetic graph/wavefront benchmark (event/count domain)

Table VI reports the three main budget-matched sequential methods. Race-fixed, the arrival-time oracle, and the fixed-time rule are summarised separately because they do not define comparable operating points in this setting. At the lowest target budget (32), interneurOn attains both the highest accuracy and the lowest composite risk. At the larger targets (48 and 64), CumGap and Poisson-LLR achieve the highest accuracies, while interneurOn is competitive and has the lowest composite risk.

Under the miss-rate constraint, Race-fixed had only a narrow feasible operating range in this regime; its nearest feasible operating point achieved mean budget 44.4, accuracy 0.941, composite risk 0.047, and median decision time 9.

TABLE VI

SYNTHETIC GRAPH/WAVEFRONT BENCHMARK AT THREE EXPLICIT EVENT-BUDGET TARGETS (ENERGY PROXY). METHODS ARE MATCHED TO THE NEAREST FEASIBLE OPERATING POINT UNDER MISS ≤ 0.10. ACH. = ACHIEVED MEAN EVENTS, ACC = ACCURACY AT DECISION, RISK = EARLY-HALT RATE + MISS RATE, MED. t = MEDIAN DECISION TIME.

	Target budget = 32				Target budget = 48				Target budget = 64			
Method	Ach.	Acc	Risk	Med. t	Ach.	Acc	Risk	Med. t	Ach.	Acc	Risk	Med. t
interneurOn	31.8	**0.947**	**0.013**	10.0	50.1	0.934	**0.003**	13.0	64.4	0.900	**0.016**	15.0
CumGap	31.8	0.919	0.056	9.0	48.8	**0.950**	0.041	11.0	72.3	0.903	0.084	12.0
Poisson-LLR	31.8	0.919	0.056	9.0	52.1	**0.941**	0.044	11.0	60.8	**0.928**	0.066	12.0

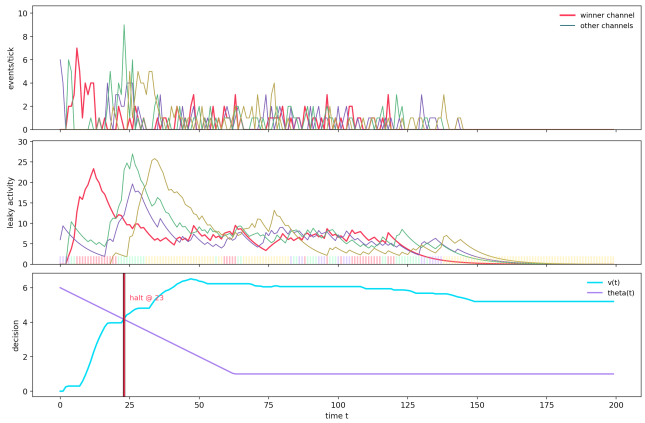

Fig. 6. Representative graph/wavefront sink trajectory: event counts at each time step from multiple sources update the leaky evidence state, and interneurOn halts once the decision variable v_t crosses the collapsing bound θ_t (here at t=23).

As additional reference points, the arrival-time oracle achieves Acc = 0.975 with zero risk, while the fixed-time rule achieves Acc = 0.178 and Risk = 1.0.

Fig. 6 shows a representative graph trial in which halting occurs once the accumulated top-2 margin drives v_t across θ_t.

C. Ablations

We probe the stabilisation mechanisms on long synthetic stress-test streams derived from INSHEP evidence. A single global gain is first calibrated on the original 20-step windows to the nominal operating point $t_{50} = 12$ ($g = 0.042376$), and this gain is then held fixed for all 200-step ablations.

Each 200-step stress-test stream combines unstable winner dynamics with frequent flips, replayed evidence segments with stronger support, and low-activity intervals near zero evidence. This tests whether the stabilisation mechanisms prevent premature commitment, chasing transient winners, and failure to re-engage after weak or conflicting evidence.

With all mechanisms enabled (leak, dwell, lock-out, lock-on, negative reset), interneurOn achieves $t_{50} = 45$ (IQR 44–47), accuracy 0.73, early-halting fraction 1.00.

Removing mechanisms:

- no dwell: $t_{50} = 43$ (IQR 25–45), accuracy 0.54;
- no lock-out: $t_{50} = 20$ (IQR 16–25), accuracy 0.17;
- no lock-on: $t_{50} = 44$ (IQR 41–45), accuracy 0.63;

- no negative reset: $t_{50} = 200$ (IQR 200–200), early-halting fraction ≈ 0.23.

These ablations show that the stabilisation mechanisms play distinct roles rather than providing redundant regularisation. Dwell mainly suppresses premature commitment, lock-out is critical for avoiding rapid winner chasing under unstable evidence, and lock-on improves label stability once a candidate hypothesis has emerged. Negative reset has the strongest effect on recoverability: without it, the decision state can remain trapped after contradictory evidence, sharply reducing the fraction of streams that halt early.

D. Per-stream decision dynamics

To illustrate collapsing-bound behaviour on individual streams, Fig. 7 shows two examples from the radar test set. Each panel plots the membrane trajectory v_t against the collapsing bound θ_t, the top-3 class posteriors, the argmax timeline with flip markers, and a matched-budget comparison to a top-2 log-odds rule.

In these examples, interneurOn halts once a stable positive margin forms, whereas the top-2 log-odds rule either commits too early or delays until the terminal step.

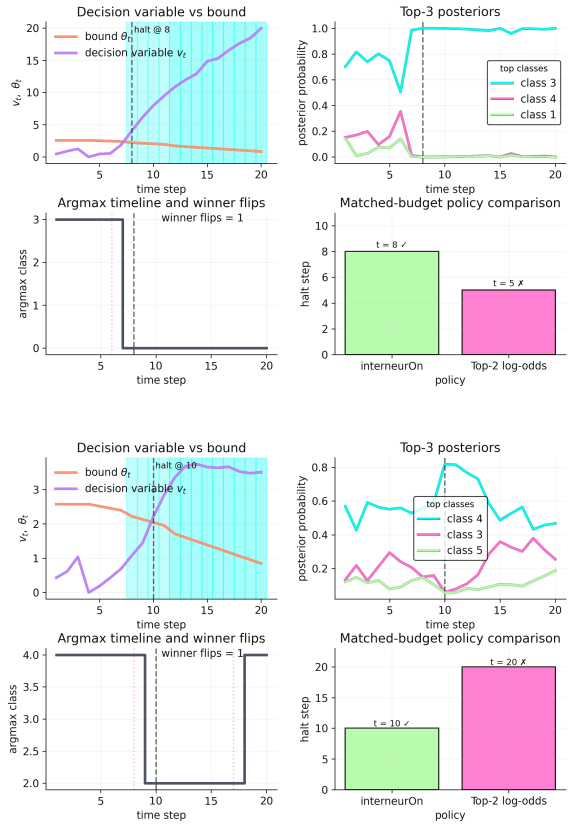

Fig. 7. **Per-stream behaviour of the collapsing-bound decision neuron.** Each panel shows one posterior stream. Top left: v_t and collapsing bound θ_t with urgency shading; the dashed line marks the halt. Top right: top-3 class posteriors. Bottom left: argmax timeline with flip markers. Bottom right: matched-budget comparison between a calibrated interneurOn and a top-2 log-odds rule. In both examples, *interneurOn* remains correct with similar or fewer time steps, whereas top-2 log-odds either stops too early and misclassifies (top) or reaches the forced halt at the terminal step (bottom).

VI. Conclusion

We presented interneurOn, a collapsing-bound decision neuron for early exit in digital neuromorphic systems. It realises stopping through leaky accumulation of a locked top-2 margin, dwell, and bound crossing, and emits a halt signal without requiring a separate controller or monitoring stage.

Across posterior-space classification experiments, a synthetic graph/wavefront benchmark, and two SpiNNaker-1 radar deployments, interneurOn remained competitive with matched early-exit baselines while preserving a hardware-compatible local stopping rule. The same mechanism transferred across calibrated posteriors, replayed class-count streams, and on-chip-generated class activity, suggesting that its behaviour depends primarily on the structure and timing of the evidence stream rather than on task-specific redesign.

These results show that early exit in neuromorphic systems need not be implemented solely as an external supervisory function, but can instead be realised as a local dynamical mechanism in the same spike/count domain as the underlying computation. This makes early exit easier to align with neuromorphic constraints such as local state, bounded per-update cost, and sparse event-driven communication.

At the same time, the results point to a practical limitation: stopping quality depends strongly on the informativeness of the incoming evidence. interneurOn performs best when upstream activity yields a sufficiently stable and graded multi-channel margin; when evidence becomes too sparse or weakly separated, reliable early stopping becomes harder even if the stopping rule itself remains unchanged.

Beyond the early-exit setting, interneurOn exploits a defining property of spiking systems: temporal structure is part of the evolving computational state, so decision-relevant information can be carried by how competition develops over time, not only by which channel is currently largest. By tracking the temporal development of the top-2 competition and converting it into a local commitment signal, interneurOn realises a spiking decision primitive that goes beyond static winner-take-all competition by representing persistence, instability, and commitment directly in its dynamics.

Natural extensions include mixed-circuit implementations, coordination of multiple local halt signals across layers or cores, and co-design of upstream encoders and local stopping circuits to make early activity more decision-informative under hardware constraints.

VII. Acknowledgment

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